funds raised in each vintage year and IRR and multiple performance measures. This should not be a surprise: exits in buyout funds depend on hot public markets. IPOs are issued in waves, which peak at the top of markets. Merger and acquisition activity is likewise strongly pro-cyclical.[19]

Biases in Private Equity Returns

Academic studies take into account two serious biases in PE data: (i) the infrequent observations of fund or company values and (ii) selection bias.[20] The latter refers to the tendency of more successful firms and funds to be overrepresented in data, causing PE investment returns to appear better than they really are. Transactions are observed only if portfolio company values are high—only if the company is doing well will it receive the next round of financing, only if the valuation is high will the company IPO, and so on—and companies not doing well (*zombies*) are often left to linger as shells without mark-to-market valuations.

Zombie valuation is often written into PE contracts (see below), by statements like "Investments will be valued at cost unless a material change justifies a different valuation."[21] As one valuation expert says, "The manager . . . has a clear incentive to hold onto investments that have poor prospects for improvement, simply to get more out of the fund in the form of management fees."[22] Zombies account for more than 5% of PE investments, according to TorreyCove Capital Partners, a consulting firm. Another survey conducted by Coller Capital, a PE investments firm, in 2011 finds that 57% of North American LPs have zombie funds in their portfolios.[23]

Academics tackling biases due to infrequent sampling and selection have used two main approaches, but both have drawbacks. First, we can examine individual portfolio company-level information. This provides more data, so there are actually returns (at irregular intervals for different companies), but selection bias is a major issue. This analysis also reflects total returns earned by a PE fund before fees and not the returns earned by an LP net of fees. Second, we can analyze PE fund-level data. The selection bias is smaller because companies that ultimately turn into zombies are eventually reflected in the fund's final closing payout. The main disadvantage of using fund-level cash flows is that these are not direct measures of returns.

[19] These are large literatures. See Rhodes-Kropf, Robinson, and Viswanathan (2005) for merger and acquisition waves and Ritter and Welch (2002) for IPO waves. Robinson and Sensoy (2011b) and Phalippou (2011) give overviews of buyout and VC cycles.

[20] There is also reporting and survivorship bias. Better-performing funds are more likely to report their performance. Phalippou and Zollo (2005) show that taking into account selection bias reduces IRRs by approximately 3%. Chapter 13 covers all these biases in more detail.

[21] Quoted from Phalippou (2009).

[22] The quotation and the estimate are from Pulliam, S., and J. Eaglesham, "Investor Hazard: 'Zombie Funds,'" *Wall Street Journal*, May 31, 2012.

[23] Coller Capital, Global Private Equity Barometer, Winter 2011–2012.